

9:00 to 10:00 Classes

Robotics, Nvidia Attach, Power and Performance advantages and design productivity

Supplier: Altera

Altera 3.0 represents the next iteration of the company making it the only pure-play FPGA company in the industry today. Learn about our value proposition in Robotics, Holoscan capabilities, and our investments into our architecture and tools you need to accelerate your design flow and beat your competition to market. Our Hyperflex and Tensor Flow DSP architecture allows for higher performance, lower power, and AI at the edge. This enables you to future proof your designs with edge AI and increase your productivity with higher utilization and ease of timing closure.

Driving Innovation with High Accuracy, Low Power RTC modules

Supplier: Micro Crystal

In today's connected products, timing is everything. From industrial automation and precision agriculture to aerospace and remote infrastructure, engineers face growing challenges around power consumption, accuracy, size, and reliability. This session explores how ultra-low-power RTC solutions can deliver precise timekeeping for years while operating in harsh environments and space-constrained designs.

Learn how Micro Crystal's high-accuracy, temperature-compensated RTC modules help extend battery life, improve system reliability, and simplify design integration. Through real-world application examples, you'll gain practical insights into solving timing challenges in next-generation embedded systems and connected devices.

Migrating from Linux: ST Enables Efficient GUI Solutions

Supplier: STMicro

In this hands-on workshop, you'll discover how ST microcontrollers and TouchGFX work together to create rich, responsive graphical user experiences using remarkably little memory. By eliminating the need for external DRAM and avoiding the overhead of Linux-based systems, you can build simpler, more efficient products.

You'll also explore partial refresh technology—a breakthrough approach that dramatically reduces memory requirements while making system design even more elegant.

We'll provide everything you need to get started: ST NUCLEO development kits, preconfigured laptops, and a guided experience that takes you from concept to a working graphical interface.”

Audio Component Selection & Acoustic Design for Modern Electronic Products

Supplier: Raltron

This session introduces the fundamentals of selecting and integrating audio components into modern electronic designs. Participants will explore the differences between speakers, piezo and electromagnetic transducers/buzzers, MEMS and electret condenser microphones, and piezoelectric elements, with an emphasis on choosing the appropriate technology for the application.

The discussion will cover key engineering considerations including SPL, frequency response and resonance, impedance, drive voltage and power, sensitivity, acoustic chambers, sound ports, environmental protection, and component size. Practical examples will demonstrate how enclosure design, mounting, drive conditions, and measurement distance can significantly affect real-world acoustic performance.

Attendees will also learn how custom acoustic chambers, 3D prototyping, lead wires/connectors, mesh, gaskets, and IP-rated constructions can be used to optimize performance when standard components do not meet the application requirements. Examples from medical, industrial, fire & safety, IoT, wearables, automotive, and consumer electronics will be used throughout the presentation to illustrate common design challenges and solutions.

10:15 to 11:15 Classes

Beyond the Hardware: Future-Ready Strategies for Industrial Computing

Supplier: Advantech

Today's industrial computing landscape is evolving rapidly with AI, edge computing, and ongoing supply chain challenges. This session explores the key technology and market trends shaping the future, along with practical strategies for building resilient, future-ready solutions. Attendees will gain valuable insights to help navigate technology transitions, reduce risk, and make informed decisions with confidence.

Silicon Secrets Revealed: Fundamentals of Memory Cell Design

Supplier: Infineon

Ever wondered what's really happening inside a flash memory device? Go beyond the datasheet and explore how non-volatile memory cells are built, programmed, and scaled at the silicon level. This session breaks down the physics behind data storage and reveals the key architectural differences between NAND and NOR technologies that drive performance, density, and reliability.

Whether you're designing embedded systems, industrial controls, or connected products, you'll gain a deeper understanding of the memory technologies powering today's electronics. We'll also highlight how Infineon's memory portfolio helps engineers address real-world design challenges across a wide range of applications.

Power Protection Reimagined – Leveraging eFuse Portfolio for Next-Generation System Reliability

Supplier: onsemi

As power density, system complexity, and uptime requirements continue to increase, intelligent power protection is becoming essential across industrial, automotive, and computing applications. This training explores the growing adoption of eFuses and highlights how onsemi's power protection portfolio delivers faster, more accurate fault protection, enhanced diagnostics, and improved system reliability compared to traditional protection methods.

Custom Magnetics for Optimized Circuits

Supplier: YAGEO Group

With the ever-increasing electrification of vehicles, the boom in global renewable energy production, and of course, the explosion of AI and datacenter related business activity in recent years, electronics industries have collectively seen increasing demand. While these modern industries present lucrative opportunities, they have also placed increasing pressure on electronics suppliers and engineers to innovate by pushing efficiencies, power densities, and topologies further than they have before.

This presentation will seek to give an overview of various magnetic components commonly used by electronics engineers and provide some insight into how customization can provide value and assist engineers in pushing performance and driving down costs in the face of new market requirements. The presentation will focus on inductors and transformers, but touch on chokes, sensing products, and others. The presentation will seek to elucidate the realities and constraints involved in commercial scale production of magnetic components while also explaining the performance and cost benefits that can be achieved through customization. The presentation will also include numerous specific examples of custom magnetics for various applications to illustrate to the audience the breadth of what can potentially be unlocked through customization of magnetic components. Through this presentation, Yageo Group hopes to inform listeners of the tremendous potential for customized magnetics to enable the circuits of the future.

1:15 to 2:15 Classes

Power Supply System EMI Fundamentals

Supplier: Advanced Energy

This session introduces the fundamentals of electromagnetic interference (EMI) in modern power electronic systems, where higher switching frequencies and power densities increase EMI challenges. Participants will learn how EMI is generated, the roles of electric and magnetic fields, and the importance of mitigation for product performance, compliance, and reliability. The course covers conducted and radiated emissions in power supplies, dominant noise paths in switching converters, and disturbance propagation mechanisms. It also reviews key EMI standards, testing methods, and compliance requirements for industrial, medical, and commercial products. Finally, attendees will explore Advanced Energy's EMI laboratories, engineering expertise, and customer support resources.

Innovation Driven by Automotive, Designed for Rugged Applications

Supplier: Molex

The challenges facing today's automotive industry—higher power, faster data, smaller packaging, and uncompromising reliability—are accelerating innovations that extend well beyond passenger vehicles. As electrification, software-defined architectures, automotive Ethernet, and advanced sensing become mainstream, connector technologies are evolving to deliver greater performance in increasingly demanding environments.

The seminar will showcase a broad range of Molex automotive technologies that are increasingly finding applications beyond passenger vehicles. Whether you're designing commercial vehicles, industrial equipment, agricultural machinery, robotics, marine systems, or other rugged applications, you'll discover how technologies developed for the automotive industry's demanding requirements can help improve reliability, performance, and enable scalable architectures to prepare your products for the next generation of connectivity.

SimSurfing: Faster Passive Component Selection: More Time for Application Design

Supplier: Murata

In this session, Murata will show how to use their free web-based simulation tool, SimSurfing, which gives engineers detailed characteristic data to run accurate simulations under specific operating and design requirements. This allows engineers to quickly select components, finishing the passive portion of their design faster and freeing up time for other design considerations. Attendees will be able to select and MLCC/EI kit to be delivered later, free of charge.

Architecting EdgeAI Systems with the i.MX 95: A Deep Dive for Embedded Engineers

Supplier: NXP

This session provides a technical deep dive into NXP's i.MX 95 applications processor family, focusing on system architecture, subsystem behavior, and design considerations for AI enabled embedded platforms. Engineers will learn how to evaluate the i.MX 95's capabilities, work effectively with its heterogeneous processing elements (and accelerators), and integrate its AI/ML features into production grade designs across industrial automation, automotive edge, medical, commercial IoT, and connected embedded applications.

The session also shows how the i.MX 95 fits into NXP's broader secure and scalable EdgeAI portfolio, from ultra low power microcontrollers to high performance application processors and discrete NPUs delivering up to 40 eTOPS, offering a unified path for deploying intelligent systems across diverse performance and power requirements.

From Edge to Cloud: Unlocking Industrial IoT with Wi-Fi HaLow

Supplier: Vantron

Wi-Fi HaLow is redefining Industrial IoT connectivity with kilometer-scale range, low power consumption, strong penetration, and IP-native networking. In this session, see how Wi-Fi HaLow-enabled devices, routers, and NXP i.MX94 gateways work together to connect remote sensors, edge AI, and even video transport applications across large industrial environments where traditional wireless technologies fall short.

Explore practical deployment architectures, key Wi-Fi HaLow design advantages, and real-world use cases from Vantron that demonstrate how long-range wireless connectivity can enable scalable, secure, and cost-effective Industrial IoT networks for energy, infrastructure, agriculture, and other remote applications.

2:30 to 3:30 Classes

How to Scale Edge AI from Development to Deployment

Supplier: Digi

Moving AI from the lab to the edge is often harder than building the model itself. Hardware complexity, diverse accelerators, and deployment challenges can slow innovation and limit scalability. In this session, see how edge AI and cloud services work together to simplify deployment, monitoring, retraining, and over-the-air updates across the full AI lifecycle.

Using a real-world optical inspection application from food production, we'll demonstrate a practical approach to building scalable, continuously improving edge AI systems, helping engineers accelerate deployment while reducing operational complexity.

Rotary Position Sensors and Technology in Aerospace Defense, and Industrial Applications

Supplier: Honeywell

Discover the latest advances in rotary position sensing and how they're enabling the next generation of aerospace, defense, industrial, and autonomous systems. Honeywell Sensing Solutions will explore the technologies behind high-reliability position feedback, from contactless Hall Effect and magnetoresistive sensors to proven magnetic and resistive solutions, along with integrated sensing architectures that improve system performance and durability.

See what's next in rotary sensing as Honeywell discusses emerging innovations designed for drones, collaborative combat aircraft, and other SWaP-C constrained platforms. Learn how digital interfaces, communication bus integration, and advanced sensor technologies are shaping tomorrow's motion control and position-sensing designs.

Discover a Faster Way to Develop with MPLAB® in VS Code®

Supplier: Microchip

See how Microchip's MPLAB Extensions bring embedded development directly into Visual Studio Code, combining a familiar IDE experience with powerful debugging and build tools. This session walks through creating, building, and debugging projects while highlighting tips to streamline your workflow. You'll also get a first look at the MPLAB AI Coding Assistant and how it can help accelerate development, reduce repetitive tasks, and improve productivity throughout the design cycle.

Powering the Next Chapter

Supplier: RECOM

Explore the evolving technologies shaping power conversion across industrial, medical, and household applications. This seminar will examine key trends in AC/DC and DC/DC power supplies, including the growing demand for higher efficiency, greater integration, enhanced protection, and application-specific solutions. We'll discuss advancements in DIN Rail power, electronic protection and monitoring, and discrete power conversion technologies. Whether optimizing for performance, space constraints, or cost, this session will provide practical insights to help engineers make informed power design decisions, improve system performance, and accelerate development.

A New Era of High-Performance Timing Solutions

Supplier: Skyworks

This session examines the latest trends in clock tree design across key industry standards including PCIe and Ethernet, and how Skyworks timing solutions help engineers achieve optimal system performance. Attendees will learn how to leverage Skyworks tools to meet the precise timing requirements of modern architectures, and explore reference design platforms spanning FPGAs, switch SoCs, embedded processors, and data center applications that utilize Skyworks timing products.